

1.3 Chemical Bonding

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.3 Chemical Bonding
Difficulty	Easy

Time allowed: 50

Score: /41

Percentage: /100

Question 1a

There are a number of different types of intermolecular forces possible between molecules.

Name the three main types and place them in ascending order of strength.

[2 marks]

Question 1b

Methanol (CH_3OH) is a small alcohol molecule with many uses.

Draw a diagram to show how two methanol molecules interact with each other through hydrogen bonding in the liquid phase.

Include all partial charges and all lone pairs of electrons in your diagram.

[2 marks]

Question 1c

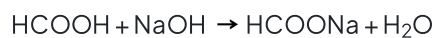
Methanol, CH_3OH can be oxidised to methanal, CH_2O and then to methanoic acid, HCOOH .

Identify the strongest type of intermolecular force between:

- | | | |
|--------------------------|--|-----|
| i) | | |
| Methanal molecules | | [1] |
| ii) | | |
| Methanoic acid molecules | | [1] |
| iii) | | |
| Water and methanal | | [1] |
| iv) | | |
| Water and methanoic acid | | [1] |

[4 marks]**Question 1d**

Methanoic acid reacts with sodium hydroxide to form sodium methanoate:

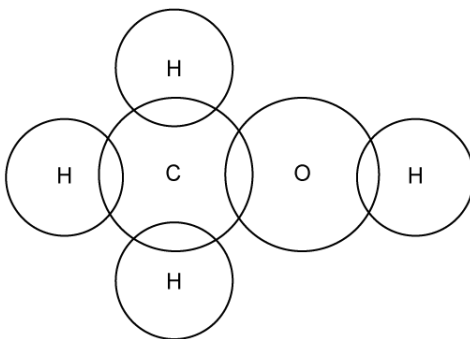


Explain why sodium methanoate is a solid at room temperature and methanoic acid is a liquid.

[3 marks]

Question 1e

Complete the dot-and-cross diagram of methanol, CH_3OH .



[2 marks]

Question 2a

Sulfur can form bonds with six fluorine atoms to form sulfur hexafluoride, SF_6 .

i)

How many electrons are in the outer shell of the sulfur in SF_6 ?

[1]

ii)

State the minimum and maximum numbers of electrons possible in the outer shell of sulfur.

[1]

[2 marks]

Question 2b

Sulfur has no lone pairs when bonded to fluorines in SF_6 .

Identify the shape of a molecule of sulfur hexafluoride, SF_6 .

[1 mark]

Question 2c

State the F-S-F bond angles in SF_6 .

[1 mark]

Question 2d

Sulfur has an electronegativity value of 2.58 and fluorine has an electronegativity value of 3.98.

i)
Suggest whether the S-F bond is polar or not.

[1]

ii)
Predict if the SF_6 molecule is polar and explain your answer.

[2]

[3 marks]

Question 3a

Sodium is a metal in the Periodic Table.

Draw a labelled diagram to show the arrangement of particles in a crystal of sodium.

In your answer, include at least six particles of each type.

[3 marks]**Question 3b**

Sodium can react with chlorine to produce sodium chloride.

Sodium chloride is ionically bonded.

State the meaning of the term 'ionic bond'.

[1 mark]**Question 3c**

Draw the 'dot and cross' diagram to show the ions formed by sodium chloride.

[2 marks]**Question 4a**

This question is about electronegativity.

Define the term electronegativity.

[2 marks]

Question 4b

A bond between nitrogen and hydrogen can be represented as $\overset{\delta-}{\text{N}} - \overset{\delta+}{\text{H}}$.

i)

In this representation, state the meaning of the symbol $\delta+$.

[1]

ii)

From this bond representation, state what can be deduced about the electronegativity of nitrogen relative to that of hydrogen.

[1]

[2 marks]

Question 4c

State **three** factors which affect the electronegativity of an element.

[3 marks]

Question 4d

i)

Identify the element in Period 4 with the highest electronegativity value.

[1]

ii)

Explain why chlorine is more electronegative than oxygen.

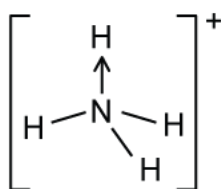
[2]

[3 marks]

Question 5a

The reaction between an ammonia molecule and a hydrogen ion forms an ammonium ion.

This ion can be represented as shown in the diagram below.



- i)
Name the type of bond represented in the diagram by N—H.

[1]

- ii)
Name the type of bond represented in the diagram by N→H

[1]

[2 marks]

Question 5b

In terms of electrons, explain why an arrow is used to represent this N→H bond.

[1 mark]

Question 5c

In terms of electron pairs, explain why the bond angles in the NH_4^+ ion are all 109.5° .

[2 marks]

Model Answers: Easy

1a

a) The three main types of intermolecular forces are:

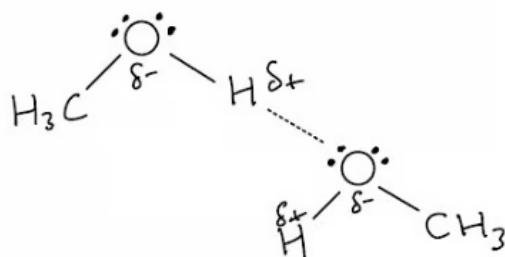
- Induced dipole-dipole interactions, permanent dipole-dipole interactions and hydrogen bonding; [1 mark]
- (In order of ascending strength) Induced dipole-dipole interactions < permanent dipole-dipole interactions < hydrogen bonding; [1 mark]

[Total: 2 marks]

- These terms are the ones that should be used, as the older term of Van der Waals is an inclusive term that covers all types of intermolecular bonding except for hydrogen bonding, and does not distinguish between induced and permanent dipole-dipole interactions
- It is essential to know the strengths in order

1b

b) A diagram to show how two methanol molecules interact with each other through hydrogen bonding in the liquid phase is:



- Two lone pairs on at least one oxygen atom
AND
On at least one OH δ^+ on H and δ^- on O; [1 mark]

- Dotted line shown between lone pair on one molecule and the correct H on another; [1 mark]

[Total: 2 marks]

- It is good practice to draw the lone pairs on every oxygen, but not essential
- Hydrogen bonding occurs between oxygen, nitrogen or fluorine atoms that are part of a polar bond, and therefore have a slight negative charge, and between hydrogens that are part of a polar bond, and have a slight positive charge
- Hydrogen bonds should not come from hydrogens that are bonded with carbons, as the C-H bond is not significantly polar

1c

c) The strongest types of intermolecular forces between each set of molecules are:

i) In methanal:

- permanent dipole-dipole interactions; [1 mark]

ii) In methanoic acid:

- hydrogen bonding; [1 mark]

iii) Between water and methanal:

- hydrogen bonding; [1 mark]

iv) Between water and methanoic acid:

- hydrogen bonding/ ion-dipole forces; [1 mark]

[Total: 4 marks]

- Whilst all of the molecules have a polar bond to an oxygen, and therefore an oxygen with a partial negative charge, methanal does not have any polar bonds to hydrogens and so cannot form hydrogen bonds to other methanal molecules
- All of the molecules can form hydrogen bonds with water as water has polar bonds to both hydrogen and oxygen, allowing hydrogen bonds to form with the oxygens on all of the organic molecules and the O-H hydrogens in methanol and methanoic acid
- Methanoic acid partially ionises, as it is a weak acid, so also forms ion-dipole

interactions with water

1d

d) Sodium methanoate is a solid at room temperature, and methanoic acid is a liquid because:

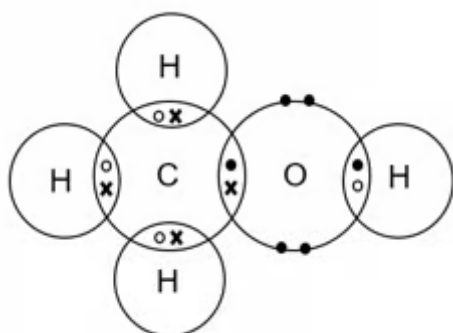
- Sodium methanoate is ionic/ contains Na^+ and HCOO^- / is an ionic lattice; [1 mark]
- Methanoic acid is a simple covalent substance/ is a molecule/ has hydrogen bonds between molecules; [1 mark]
- Ionic bonds are stronger/ require more energy to break/ require a higher temperature to break than hydrogen bonds ORA; [1 mark]

[Total: 3 marks]

- This reaction is an example of an acid base reaction using an organic acid, so it still produces a salt and water
- Salts are ionic, and so ionic bonding occurs in that structure and the usual properties of ionic substances apply
- Methanoic acid has the strongest type of intermolecular force present, hydrogen bonding, but this is still not close to as strong as ionic bonds are, so the melting point is much lower

1e

e) The completed dot-and-cross diagram of methanol is:



- Five bonding pairs of electrons between carbon, oxygen and hydrogen; [1 mark]
- Two lone pairs of electrons on oxygen; [1 mark]

[Total: 2 marks]

- When drawing dot-and-cross diagrams you only need to draw the outer shell electrons
- Make sure that you check that the atoms all have full outer shells
- Any combination of dots and crosses can be used, as long as the electrons are consistent for each atom

2a

a)

i) The number of electrons in the outer shell of sulfur in SF_6 is:

- Twelve; [1 mark]

ii) The maximum and minimum electrons possible in the outer shell of sulfur are:

- Minimum of 8 (electrons)

AND

Maximum of 18 (electrons); [1 mark]

[Total: 2 marks]

- Sulfur is in Period 3, so it can use the empty d-orbitals to allow more electrons than the standard octet, up to a maximum of 18 to form an expanded octet

2b

b) The shape of a molecule of SF_6 is:

- Octahedral; [1 mark]

[Total: 1 mark]

- There are six electron domains around the sulfur, one single bond with each fluorine, which means an octahedral shape for the molecule

2c

c) The F-S-F bond angles in SF_6 are:

- 90° ; [1 mark]

[Total: 1 mark]

- All bond angles in octahedral molecules are 90° , as this is the furthest away from each other that the electron domains can get before they are getting closer again!

2d

d)

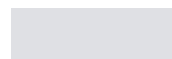
i) Predicting the polarity of the S-F bond:

- (The S-F bond) is polar; [1 mark]

ii) Predicting the polarity of SF_6 :

- SF_6 is not polar/ is a non-polar molecule; [1 mark]
- As it has no overall dipole/ the distribution of electrons is even across the molecule as a whole; [1 mark]

[Total: 3 marks]



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2d

d)

i) Predicting the polarity of the S-F bond:

- (The S-F bond) is polar; [1 mark]

ii) Predicting the polarity of SF₆:

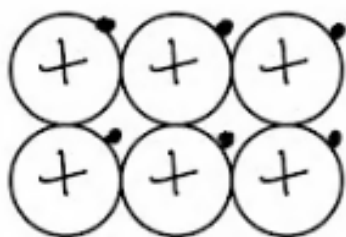
- SF₆ is not polar/ is a non-polar molecule; [1 mark]
- As it has no overall dipole/ the distribution of electrons is even across the molecule as a whole; [1 mark]

[Total: 3 marks]

- Bonds are polar when there is a sufficient difference in electronegativity, generally treated as a difference of 0.5 or more between the values for the different atoms
- Fluorine is the most electronegative element that can form bonds and forms polar bonds with all other elements except itself
- Molecules are polar when they have an uneven distribution of electrons across the molecule in any direction; this can be the result of unsymmetrical polar bonds or non-polar bonds

3a

a) A labelled diagram to show the arrangement of particles in a crystal of sodium is below:



- A regular arrangement of metal ions; [1 mark]
- + sign drawn in each metal ion; [1 mark]
- At least six electrons drawn; [1 mark]

[Total: 3 marks]

- In a metal the outer electrons of each atom become delocalised leaving them to form ions with a positive charge
- You therefore **must** show the positive charge in each ion
- The question specifically states to draw six ions, so if you have drawn less you will not score marking point 1
- Make sure you draw the ions in a regular arrangement to show that the ions can slide over each other with ease
- The electrons do not need to have a negative charge for you to score the mark

3b

b) The definition of ionic bonding is:

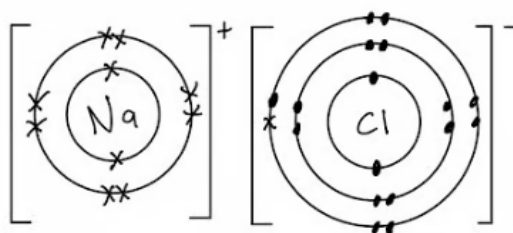
- Electrostatic attraction between oppositely charged ions (in a lattice); [1 mark]

[Total: 1 mark]

- The metal loses electrons to form positively charged ions (cations)
- The non-metal gains electrons to form negatively charged ions (anions)
- The opposite charges are strongly attracted

3c

c) The ions formed by sodium chloride are:



- Electronic configuration of sodium is 2,8

AND

Ion has a single positive charge; [1 mark]

- Electronic configuration of chloride ion is 2,8,8

AND

Ion has a single negative charge; [1 mark]

[Total: 2 marks]

- You must show all the electrons in ionic dot and cross diagrams
- **Remember:** Sodium will lose its outer electron to achieve a full outer shell
- It has one more proton than electron overall so the ion has a single positive charge
- Chlorine gains the one electron, meaning the chloride ion has one more electron than proton and therefore has a single negative charge
- It is good practice to use dots for electrons in one ion and the other with crosses however you would score the mark if you have done either as long as the number of them is correct

4a

a) The term electronegativity means:

- The ability of an atom / nucleus to attract electrons to itself; [1 mark]
- In a covalent bond; [1 mark]

[Total: 2 marks]

- This is a common exam question so you **must** learn this definition!

4b

b)

i) The meaning of the symbol δ^+ is:

- Electron deficient; [1 mark]

ii) From this bond representation it can be deduced that:

- Nitrogen is more electronegative than hydrogen; [1 mark]

[Total: 2 marks]

- The more electronegative atom has a partial negative charge because it draws the bonding pair of electrons towards itself
- The less electronegative atom has a partial positive charge as the pair of electrons are drawn away from it
- You will also score the mark for b) i) for stating that the symbol means there is a partial negative charge

4c

c) Three factors which affect the electronegativity of an element are:

- Nuclear charge; [1 mark]
- Atomic radius; [1 mark]
- Shielding (by inner electrons); [1 mark]

[Total: 3 marks]

- **Remember:** The more electronegative an element, the more it will attract electrons to itself
- Nuclear charge refers to the number of protons in the nucleus, the more protons the more attraction there will be between them and the negative electrons
- Atomic radius is the distance between the nucleus and electrons in the outer shell - the further away the electrons are, the harder it will be to attract them
- Shielding of outer electrons by inner electrons is affected by the number of shells an atom has - the more shells, the greater the effect of shielding and the harder it is to attract electrons

4d

d)

i) The element in Period 4 with the highest electronegativity value is:

- Bromine; [1 mark]

ii) Chlorine is more electronegative than oxygen because:

- The nuclear charge increases
AND
Atomic radius decreases; [1 mark]

- (But) there is similar shielding; [1 mark]

[Total: 3 marks]

- Going across a period electronegativity increases
- This is because each element has one more proton in the nucleus (increasing nuclear charge) but the number of energy levels and therefore amount of shielding remains the same
- Electrons will be more strongly attracted to the nucleus further along period
- **Careful:** some students mistakenly identify the row with lithium as being Period 1, when Period 1 contains hydrogen and helium

5a

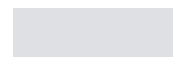
a)

i) The type of bond represented in the diagram by N—H is:

- Covalent; [1 mark]

ii) The type of bond represented in the diagram by N→H is:

- Dative covalent / co-ordinate; [1 mark]



- (But) there is similar shielding; [1 mark]

[Total: 3 marks]

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5a

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- Covalent; [1 mark]

ii) The type of bond represented in the diagram by N→H is:

- Dative covalent / co-ordinate; [1 mark]

[Total: 2 marks]

- **Remember:** A single covalent bond is one shared pair of electrons with one electron donated from each atom
- A dative covalent or co-ordinate bond is one in which one atom donates both the electrons in the pair to another atom

5b

b) An arrow is used to represent this N→H bond because:

- It shows that both / two / pair of electrons come from nitrogen; [1 mark]

[Total: 1 mark]

- The hydrogen is electron deficient due to there being a single positive charge on the ion

- The nitrogen atom has one lone pair of electrons so donates these electrons to the hydrogen so that it can achieve a full outer shell
- The arrow shows the direction in which the electrons are being donated

5c

c) The bond angles in the NH_4^+ ion are all 109.5° because:

- There are four bonding pairs of electrons; [1 mark]
- Which repel each other equally; [1 mark]

[Total: 2 marks]

- The molecule will adopt a shape which minimises the repulsive forces of the bonding pairs of electrons
- When explaining the bond angles and shape for a molecule, always count the number of bonding and lone pairs of electrons
- **Remember:** The order of repulsion is:
 - lone pair – lone pair > lone pair – bond pair > bond pair – bond pair
- In this case, there are only bonding pairs of electrons so they will repel each other equally to give the molecule a tetrahedral shape